

Case Study — Telecom Segmentation

Overview of Customer Segmentation

Customer segmentation (also known as market segmentation) is the division of potential customers in a given market into discrete groups. That division is based on customers having similar enough:

- Needs, i.e., so that a single whole product can satisfy them.
- Buying characteristics, i.e., responses to messaging, marketing channels, and sales channels, that a single go-to-market approach can be used to sell to them competitively and economically.

There are three main approaches to market segmentation:

- A priori segmentation — the simplest approach, uses a classification scheme based on publicly available characteristics (such as industry and company size) to create distinct groups of customers within a market. However, a priori segmentation may not always be valid, since companies in the same industry and of the same size may have very different needs.
- Needs-based segmentation — based on differentiated, validated drivers (needs) that customers express for a specific product or service. Needs are discovered and verified through primary market research, and segments are demarcated based on those needs rather than characteristics like industry or company size.
- Value-based segmentation — differentiates customers by their economic value, grouping customers with the same value level into segments that can be distinctly targeted.

Benefits of Customer Segmentation

Without a deep understanding of how a company's best current customers are segmented, a business often lacks the market focus needed to allocate its human and capital resources efficiently. A lack of segment focus can cause diffused go-to-market and product strategies, ultimately impeding growth.

Done right, segmentation can tangibly impact operating results by:

- Improving the whole product — knowing who buys and why helps differentiate the company as the best solution for individual needs, extending to better support, services, and overall customer experience.
- Focusing marketing messages — segment-specific messaging results in higher quality inbound interest.
- Allowing sales to pursue higher-percentage opportunities — less time on low-value prospects, more time on the most successful segments, increasing win rate and revenue.
- Getting higher quality revenues — selling into the wrong segment can be costlier to serve, with higher churn and lower upsell potential; focusing on the right segments improves margins and stability.

Business Problem

One of the leading telecom companies would like to define marketing strategy by analyzing their customers' behavior. The company provided a sample dataset which summarizes the behaviors of active customers of a mobile network operator.

Expectations from the Trainees

- Identification of the relationships/affinities between services.
- Clustering: Apply a data reduction technique (factor analysis) for variable reduction, and a clustering algorithm to reveal the behavioural segments of customers.
- Identify cluster characteristics using detailed profiling.
- Provide strategic insights and implementation strategies for the given set of cluster characteristics.

Data Available

Telco Data.csv

Field	Description
Region	Geographic indicator
Tenure	Months with service
Age	Age in years
Marital	Marital status
Address	Years at current address
Income	Household income in thousands
Ed	Level of education
Employ	Years with current employer
Retire	Retired
Gender	Gender
Reside	Number of people in household
Tollfree	Toll free service (1 = Using, 0 = Not Using)
Equip	Equipment rental (1 = Using, 0 = Not Using)
Callcard	Calling card service (1 = Using, 0 = Not Using)
Wireless	Wireless service (1 = Using, 0 = Not Using)
Longmon	Long distance last month usage

Tollmon	Toll free last month usage
Equipmon	Equipment last month usage
Cardmon	Calling card last month usage
Wiremon	Wireless last month usage
Multline	Multiple lines (1 = Using, 0 = Not Using)
Voice	Voice mail (1 = Using, 0 = Not Using)
Pager	Paging service (1 = Using, 0 = Not Using)
Internet	Internet (1 = Using, 0 = Not Using)
Callid	Caller ID (1 = Using, 0 = Not Using)
Callwait	Call waiting (1 = Using, 0 = Not Using)
Forward	Call forwarding (1 = Using, 0 = Not Using)
Confer	3-way calling (1 = Using, 0 = Not Using)
Ebill	Electronic billing (1 = Using, 0 = Not Using)
Custcat	Customer category